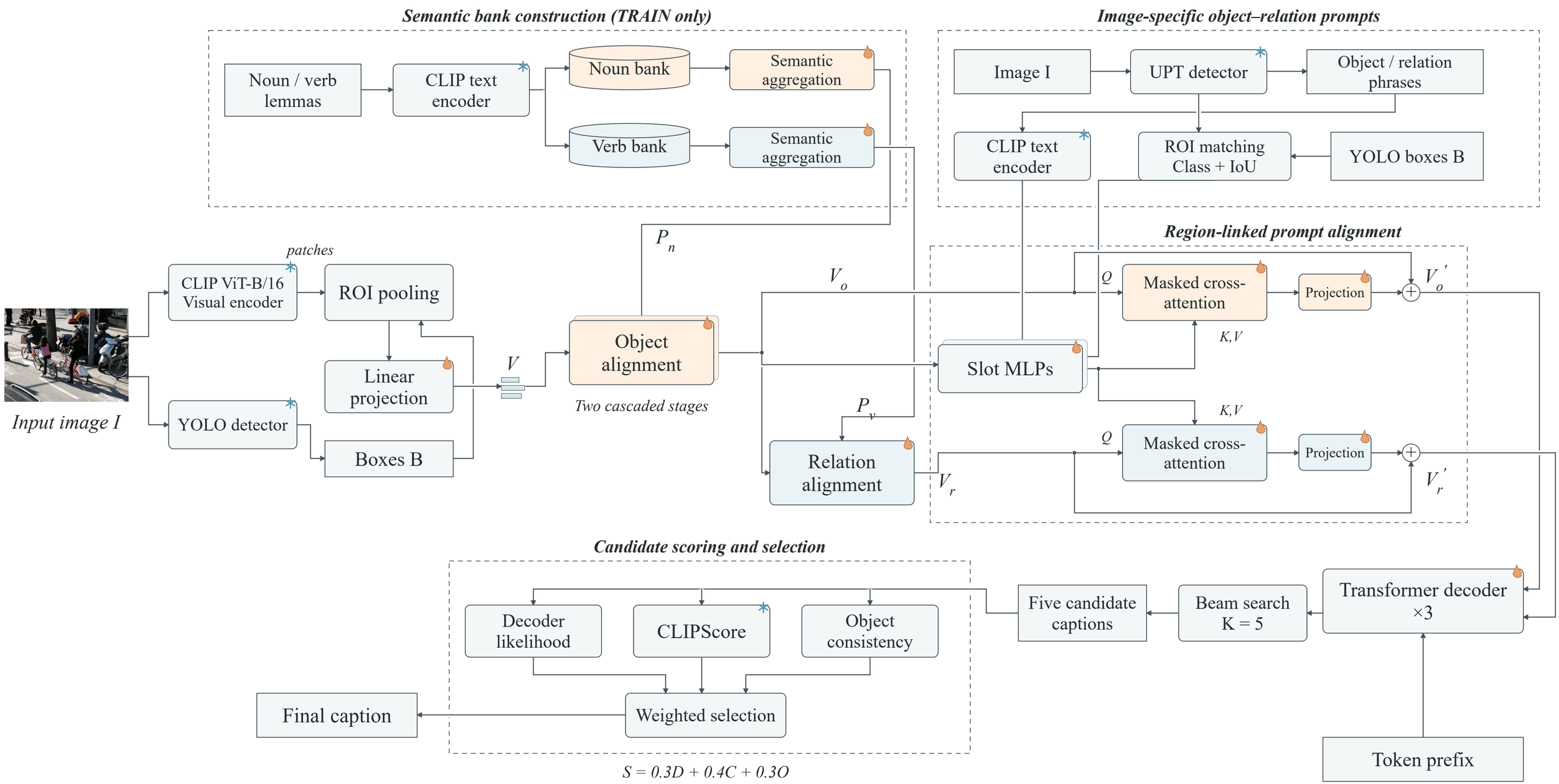

Semantic bank construction (TRAIN only)
Noun / verb lemmas
CLIP text encoder
Noun bank
Verb bank
Semantic aggregation
Semantic aggregation
Image-specific object–relation prompts
Image I
UPT detector
Object / relation phrases
CLIP text encoder
ROI matching Class + IoU
YOLO boxes B
Region-linked prompt alignment
patches
CLIP ViT-B/16 Visual encoder
ROI pooling
Linear projection
YOLO detector
Boxes B
Input image I
V
Object alignment
Two cascaded stages
P_n
V_o
P_v
Relation alignment
V_r
Slot MLPs
Q
Masked cross-attention
Projection
V_o'
K,V
K,V
Q
Masked cross-attention
Projection
V_r'
Candidate scoring and selection
Decoder likelihood
CLIPScore
Object consistency
Weighted selection
Final caption
S = 0.3D + 0.4C + 0.3O
Five candidate captions
Beam search K = 5
Transformer decoder ×3
Token prefix